

Santiago Páez Avendaño

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EDUCATION

M.Sc. student in Astrophysics

Instituto de Astronomía, Universidad Nacional Autónoma de México (UNAM)

Mexico City, Mexico

Aug. 2024 – Expected Aug. 2026

- Advance: three out of four semesters
- GPA: 9.44/10.0

B.A. in Natural Sciences Education

Faculty of Education, Universidad de La Sabana

Chia, Colombia

Jan. 2019 – Mar. 2024

- GPA: 4.16/5.0
- Awarded thesis: “Exoplanet Hackathon: A challenge-based learning proposal for the development of scientific skills”.

RESEARCH EXPERIENCE

SCIENTIFIC ARTICLES

- Khandelwal, A. et al. [included **Páez, S.**] (2026). “TOI-6884: A Low-Mass Brown Dwarf Transiting an F-Type Star”. *Monthly Notices of the Royal Astronomy Society* In revision during Mar. 2026.
- **Páez, S.**, Gómez Maqueo Chew, Y., and Hebb, L. (2026). “Data reduction method for OPTICAM multiband time series of transiting exoplanets”. *RAS Techniques and Instruments* rzag021.
- Gómez Maqueo Chew, Y. et al. [included **Páez, S.**] (2026). “TOI-1080 b: a temperate, rocky planet orbiting a quiet M4V host”. *Monthly Notices of the Royal Astronomical Society* stag438.
- **Páez, S.**, García, I., and Acevedo, A. (2022). “Enseñanza de la expresión genética: modelo y modelización en grado décimo”. *Bio-grafía* 15(29), pp. 54–66.
- Leal Leal, M. et al. [included **Páez, S.**] (2021). “Mars Dichotomy: Prospects for human life on Mars”. *EarthArXiv eprints* X51S5J.

OBSERVATIONAL EXPERIENCE

- Gómez Maqueo Chew, Y. et al. [included **Páez, S.**] Insights into stellar activity cycles: Spot crossing during transit. Time: 3 nights in Mar. 2026. Instrument: OPTICAM/OAN-SPM 2.1m Telescope. Role: Co-Investigator leading data acquisition and data reduction.
- Hebb, L. et al. [included **Páez, S.**] Mass Measurements of Two Gas Giant Planets around M-dwarf stars. Time: 1.75 nights in 2026A. Instrument: WIYN/NEID. Role: Co-Investigator involved in preparation of observing proposal and transit fitting.
- Khandelwal, A. et al. [included **Páez, S.**] Multi-band Transit Follow-up of Giant Planet Candidates Around M-Dwarfs with OPTICAM. Time: 6 nights in Feb. 2026. Instrument: OPTICAM/OAN-SPM 2.1m Telescope. Role: Co-Investigator involved in preparation of observing proposal, design of observation plan, and leading the data acquisition, data reduction, and transit fitting.
- Gómez Maqueo Chew, Y. et al. [included **Páez, S.**] Insights into stellar activity cycles: Spot crossing during transit. Time: 3 nights in Mar. 2025. Instrument: OPTICAM/OAN-SPM 2.1m Telescope. Role: Co-Investigator leading data acquisition and data reduction.
- Khandelwal, A. et al. [included **Páez, S.**] Transit follow-up observations of giant planet candidates around M-dwarfs through OPTICAM multi-band photometry. Time: 6 nights in Feb. 2025. Instrument: OPTICAM/OAN-SPM 2.1m Telescope. Role: Co-Investigator involved in preparation of observing proposal, design of observation plan, and leading the data acquisition, data reduction, and transit fitting.

CONFERENCES, MEETINGS AND POSTERS

- **Páez, S.**, Gómez Maqueo Chew, Y., and Hebb, L. Data reduction method for OPTICAM multiband time-series of transiting exoplanets. *Oral presentation in the XXII Astronomy Students Meeting (REA)*. Mexico City, Mexico. 2025.
- **Páez, S.**, Gómez Maqueo Chew, Y., and Hebb, L. Método de reducción de datos para series de tiempo multibanda con OPTICAM de exoplanetas transitantes. *Seminar of Astrophysics and Cosmology COSMOS - Instituto de Ciencias Físicas - UNAM*. Online. 2025.
- **Páez, S.**, Tovar, D., and Barreto, C. Proyéctate: a novel strategy to reach the public audience using Astronomy as a catching science topic. Oral Presentation. *Oral presentation in Communicating Astronomy with the Public - CAP 2024*. 2024.
- León-García, A. et al. [included **Páez, S.**] Follow-up of WASP-43b and WASP-107b transits through differential photometry from OAN-TNT. Poster. *LXVI National Physics Congress*. Morelia, Mexico. 2023.

INTERNSHIPS, WORKSHOPS, AND SUMMER SCHOOLS

Code/Astro Workshop

A week-long intensive training in astronomy software development. It covered the fundamental software engineering skills and best practices necessary for building sustainable open-source packages for astronomy applications, such as documentation, code profiling, unit tests, parallelization, code packaging and release.

Evanston, IL, USA

Aug. 4–8, 2025.

XV ESAOBELA

100 hours of astronomical observation using 1-meter and 10-inch telescopes, as well as 51 hours of theoretical instruction at the National Astronomical Observatory of Tonantzintla.

Puebla, Mexico

Jan. 2023 – Feb. 2023.

XXXI Scientific Summer at the OAN-SPM

Summer school featuring theoretical classes on general astrophysics topics and remote observing sessions using the 84-cm telescope at the National Astronomical Observatory of San Pedro Mártir.

Online

Jun. 2022.

Alpha-Cen Summer School

Summer School focused on developing programming skills for handling astronomical data and introducing core topics in astrophysics.

Online

Aug. 2021.

RECOGNITIONS AND SCHOLARSHIPS

National Graduate Scholarship (Master's Degree)

Recipient of fully the funded National Graduate Scholarship by the Secretariat of Science, Humanities, Technology and Innovation (SECIHTI) of Mexico

Mexico City, Mexico

Aug. 2024 – Jun. 2026

Bogota City Council

Recognition through an official council note for being a Top 50 Finalist of the Global Student Prize 2023 (Proposition No. 521 of 2023).

Bogota, Colombia

Dec. 2023

Best Saber Pro test score

Awarded for achieving the highest Saber Pro undergraduate examination results in the Natural Sciences Education program during 2022

Chia, Colombia

Oct. 2023

Global Student Prize 2023

[Top 50 Finalist](#). In recognition of exceptional achievements in the classroom and beyond. Only Colombian in the top 50 finalists out of 5,600 applicants worldwide.

Varkey and Chegg.org Foundations

Jul. 2023

FEST Scholarship

Fully funded undergraduate scholarship by Secretariat of Education of Bogotá

Bogota, Colombia

Jan. 2019 – Nov. 2024

OUTREACH

Public Talk

“Ultra-precise light curves for studying transiting exoplanets. Clavius Astronomical Center of the Universidad Iberoamericana”.

Mexico City, Mexico
Feb. 2025

Staff Noche de Las Estrellas Festival

Outreach staff in the Guillermo Haro Planetarium during the Noche de Las Estrellas Festival.

Mexico City, Mexico
Nov. 2025

Testimonial

“In search of exoplanets. The transit of a Colombian astronomer in Mexico”. A testimonial describing the experience of pursuing a Master’s degree in astrophysics at UNAM.

Journal UNAM International
Mar. 2025

Public Talk

“Mexico in the discovery and study of distant worlds”. Orizaba Planetarium.

Orizaba, Mexico
Feb. 2025

Astronomy Outreach Festivals

Organized and participated in eight astronomy outreach festivals across multiple municipalities in the Sabana Centro province. Activities included workshops, public lectures, and nighttime observations, reaching over 4,000 participants in total.

Sabana Centro Province, Colombia
May 2019 – Nov. 2023

LANGUAGES

- Spanish – Native
- English – TOEFL exam: 79 (Reading: 23, Listening: 13, Speaking: 23, Writing: 20) | Test date: Dec. 2025

SKILLS

Programming

Python Git/GitHub MATLAB HTML/CSS R Studio

Software

Jupyter Notebook AstrolImageJ Poetry UV $\text{\LaTeX}$ Stellarium Topcat VS Code

Scientific Packages

Astropy photutils Scipy juliet exoplanet pyaneti PyMC

Operating Systems

macOS Linux